

GLANDULAR EPITHELIA

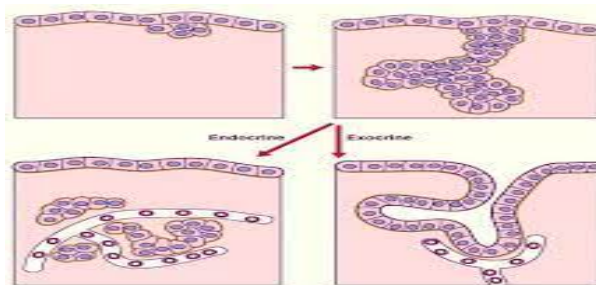
Definition

- Glandular epithelia are epithelia primarily composed of specialized cells with secretory activity: glandular cells.
- Glandular epithelium is most often located within an anatomical entity called a gland.
- Glands release their secretory product:
- Either externally via an excretory duct: This is called an exocrine gland.
- Directly into the vascular system: This is called an endocrine gland.
- Some organs, such as the liver, exhibit both endocrine and exocrine secretory activity: amphicrine gland.
- Through apposition, certain cells remain isolated and dispersed within the surface epithelium, while acquiring exocrine or endocrine secretory potential through in situ differentiation.

These are then referred to as isolated intraepithelial exocrine or endocrine cells (small intestine, trachea).

Embryology of Glandular Epithelia:

- In general, glandular epithelia form through invagination of a primitive epithelium into the underlying mesoderm. From this primitive epithelium, a solid cellular cord forms, which can develop in two directions:
- Either the cord maintains contact with the original epithelium (and therefore the exterior) and develops a lumen (future excretory duct): this is the case of an exocrine gland;
- Or it loses this contact and the cells form a solid mass. This will then be an endocrine gland.



II. Glandular Function

Regardless of the mode of secretion and the nature of the product, the glandular cell fulfills its function in three stages: the "secretory cycle":

A- The Secretion Phase

This phase can be continuous or discontinuous and consists of the uptake of metabolites from blood vessels near the connective tissue.

These metabolites are then transformed into a specific product, characteristic of the cell: the secretion.

It should be noted that the glandular cell exhibits a secretory polarity, indicated by the position of the organelles. Thus, in general: synthesis begins at the basal pole of the cell where the endoplasmic reticulum and chondriome are located, and then continues in The Golgi apparatus, most often supranuclear, marks the secretory pole of the cell. The secretory product is then accumulated at the apical pole of the cell, which constitutes the excretory pole.

B- Intracytoplasmic synthesis and storage in various forms (granules, vacuoles, etc.)

C- Excretion of the secretory product

This can occur in three main ways:

1. Merocrine secretion

Vacuoles containing the secretory product fuse with the plasma membrane and open at the surface. The cell then resumes another secretory cycle without damage (e.g., parotid gland).

2. Holocrine secretion

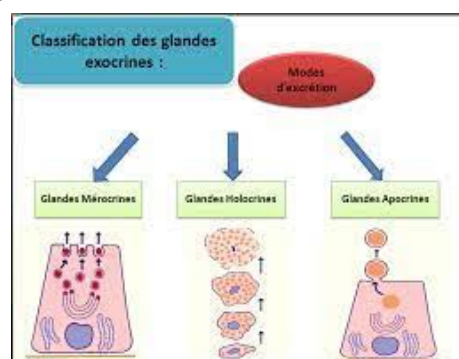
The cell undergoes only one secretory cycle. The glandular product consists of the entire cell, which is completely excreted by the gland (e.g., sebaceous gland). Basal replacement cells allow for the renewal of the gland.

This mode of secretion involves slow and continuous functioning as well as concomitant renewal.

3) Apocrine secretion

The secretory product accumulates at the apical pole of the cell, which detaches during extrusion. The glandular cell, retaining its nucleus and organelles, then begins a new secretory cycle (examples: mammary gland and some sweat glands).

Envoyer des commentaires



V. Classification of Glands

The mode of glandular excretion allows us to distinguish between endocrine and exocrine glands.

A. Exocrine Glands

➤ According to the number of glandular elements

- a. Unicellular glands: located within a lining epithelium (goblet cells in the respiratory epithelium).
- b. Intraepithelial glands: These are small groups of glandular cells embedded within a lining epithelium.
- c. Glandular organs: the glandular cells are grouped into anatomically distinct organs.

➤ According to the shape of the glands

a. Tubular glands

These are simple tubes whose distal end is composed of secretory cells and whose proximal end serves as an excretory duct.

The shape of the tubule leads to the following distinctions:

- Convoluted tubular glands
- Straight tubular glands
- Simple tubular glands consist of only one secretory duct.
- Compound tubular glands, on the other hand, are made up of several tubules opening into the same excretory duct.

b. Acinar glands

They are spherical in shape, bounded by a basement membrane upon which the secretory cells rest. These cells enclose a lumen into which the secretory product is released, and which is then drained by an excretory duct.

Simple acinar glands consist of only one secretory acinus.

Compound acinar glands are composed of several acini which, via their individual excretory ducts, empty into a common excretory duct.

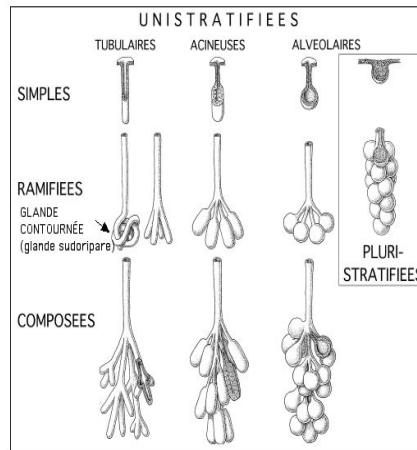
One type of acinus is the tubuloacinus, which is more elongated and oval in cross-section (exocrine pancreas).

c. Alveolar Glands

In this case, the glandular formations are less regular, sac-like and more or less bumpy. They can be simple (sebaceous gland of the skin) or compound (Meibomian gland of the eyelid).

d. Lobulated glands

Some large glands (salivary glands, pancreas) are organized into lobules, separated from each other by connective tissue. Each lobule consists of compound acinar glands drained into the lobules by intralobular ducts; these ducts drain out of the lobules by interlobular ducts.



➤ Cellular Criterion

Glandular cells have a morphological and functional polarity; they present the following characteristics: An apical pole connected to the cavity and containing the product to be excreted;

A basal pole resting on a basal lamina that separates the cell from the blood capillaries.

Among these glandular cells, we describe:

- Mucous or muciparous cells
- These are responsible for producing a viscous product: mucus.
- They are of two types:
 - Cells with a closed mucous pole (gastric cells): these constitute a protective chemical lining epithelium; these cells have an apical pole occupied by mucus and a basal pole occupied by organelles.
 - Goblet cells: these cells are scattered throughout certain epithelia.
- Tall, columnar cells shaped like a goblet, whose basal pole is occupied by organelles and whose apical pole contains mucigen globules delimited by a membrane that gradually dissociates to allow the mucigen to be released freely.
- Serous cells: parotid gland – exocrine pancreas
- These cells are responsible for producing a product rich in enzymatic proteins: zymogen granules.
- They exhibit a constantly renewing secretory cycle, divided into three phases: resting phase, loading phase, and excretion phase.
- These are pyramidal cells whose cytoplasm is divided into two regions with different staining affinities: the apical half is eosinophilic and the basal half is basophilic. The nucleus is located at the junction of two-thirds of the apical pole and one-third of the basal pole; the organelles are basal.

Note: Mucous and serous cells are frequently found in exocrine glands; they allow us to distinguish:

Purely mucous glands (laryngeal glands) or purely serous glands (parotid gland)

Mixed glands with a mucous predominance (sublingual gland) or a serous predominance (submandibular gland).

- **Sebaceous cells:** These form the sebaceous glands located throughout the skin except on the palms of the hands and the soles of the feet. They produce a lipid product called sebum, which forms a protective film on the surface of the skin.

They are of three types: basal, intermediate, and superficial. Each of these types corresponds to a stage of development.

- **Sweat cells:** These form the sweat glands, which are of two types:

Eccrine sweat glands, distributed across the epidermal surface of the body, where they release sweat.

Apocrine sweat glands, which release a lipid-rich product either into a hair follicle or at the pilosebaceous orifice. Sweat cells are arranged in two layers: a light basal layer and a dark superficial layer.

- **Lining cells:** These belong to the fundic glands of the gastric wall and produce hydrochloric acid (HCl).

- **Pulmonary epithelial cells:** These are called type II pneumocytes and produce surfactant, which is rich in phospholipids.

- **Mammary cells** line the mammary glands and produce milk, a product rich in lipids.

Endocrine Glands

They release their secretory product (or hormone) into the bloodstream and are therefore richly vascularized. Each glandular cell is in contact with a blood capillary. (Endocrine glands lack excretory ducts.)

B. a) Unicellular glands: grouped in nuclei. Ex.: Magnocellular nuclei of the hypothalamus.

B. b) Multicellular glands: several types:

Diffuse-type endocrine glands. Ex.: Endocrine pancreas.

- ✓ Endocrine glands with a vesicular architecture: e.g., the thyroid gland, where the cells are arranged in a single layer to form small spheres or vesicles. This arrangement allows for extracellular storage of the secretory product during the loading phase. At the periphery of the vesicles, the glandular cells are separated by a basal lamina from a richly vascularized connective tissue.

- ✓ Endocrine glands with a trabecular architecture: Where the glandular cells are grouped into trabeculae between which numerous blood capillaries pass. Example: adrenal cortex.

- **Mixed Glands**

Some anatomically distinct glands possess both endocrine and exocrine structures.

These glands can be:

Heterotypic : These are glands where two types of glandular cells coexist, each with distinct functions (e.g., the pancreas). The pancreas is an exocrine gland that produces pancreatic juice, which is released into the lumen of the digestive tract through excretory ducts. In addition to this function, it also contains endocrine glandular formations responsible for releasing hormones such as insulin and glucagon into the bloodstream.

omotypic: In this case, the glands are composed of cells with a dual function (e.g., liver cells).

Regulation of Exocrine Glandular Activity

Regulatory mechanisms can affect either synthesis, excretion, or both.

Control is twofold:

- ✓ **Nervous control** via the autonomic nervous system.
- ✓ **Hormonal control**: specific hormonal influence.

VI. Regulation of Endocrine Glandular Activity

This is very strict:

- ✓ Nervous (e.g., the adrenal gland)
- ✓ Endocrine (hypothalamic-pituitary system)
- ✓ Or direct (serum calcium is autoregulated by modifying parathyroid hormone secretion).